

Airbus A380

THE AIRBUS A380 WAS CONCEIVED in the early 1990s to compete with, and if possible replace, the Boeing

747, and in 1994 work began in earnest on what was then called the A3XX. The main contractors for the A380

are in France, Germany, Britain, and Spain. A custom-built ship carries fuselage sections from

Hamburg and Brittany, wings from north Wales, and tailcones from Spain to the French port of Bordeaux, from where they are transported by barge and road to the assembly plant in Toulouse.

The A380's shape is subtly molded to minimize drag from its ovoid fuselage. The structure makes extensive use of composite materials, such as thermoplastics and GLARE (aluminum and fiberglass). Its four huge engines, the most powerful ever used on an airliner, are surprisingly quiet. When carrying 550 passengers, Airbus claims the superjumbo is eco-friendly and cheap to run, since it uses only $\frac{3}{5}$ gallon (2.9 liters) of fuel per passenger per 60 miles (100km). With 22 wheels to spread its weight the A380 was compatible with most existing runways at major hubs, but its massive wingspan necessitated widening of taxiways. Airports also had to upgrade facilities for passenger deboarding and baggage handling. The first A380 entered service in 2007. Airbus delivered its 100th A380 in 2013 and by the end of 2015 the aircraft had carried over 100 million passengers.



NOSE SECTION

The lower forward part of the A380 houses the nose landing gear in flight. It is unpressurized, and is separated from the pressurized cabin and flight deck by a bulkhead. There is a strengthened bird-impact shield below the cockpit windows.

Specifications A380

ENGINE	4 x 70,000lb (311,000kg) Rolls-Royce Trent 970 or Engine Alliance GP7270
WINGSPAN	261ft 10in (79.8m)
TOTAL LENGTH	239ft 2in (72.9m)
WEIGHT (EMPTY)	610,680lb (277,000kg)
CREW	2
PASSENGERS	555 (in typical three-class arrangement)
CRUISING SPEED	575mph (927kph) (Mach 0.85)

Tall vertical tailplane

Split rudder for yaw control

MASSIVE FUSELAGE

The Airbus A380 is the world's largest airliner. Its massive fuselage incorporates three decks – upper and main decks for passengers and a lower cargo deck. With 49 percent more floor space than the Boeing 747, the A380 could theoretically carry up to 853 passengers.

Main deck windows

Jupp-Reeselet winglet

Belly fairing

